Attention-Driven Multi-Scale Representation Learning for Document Intelligence

Dr. Elena Vance, Dr. Marcus Thorne, Sophia Lin
Department of Computer Science, Institute of Advanced Technology

ABSTRACT

Academic document understanding requires joint reasoning over dense typographic layouts and semantic text. Existing vision-language models frequently suffer from spatial misalignment and high computational complexity. In this work, we present DocScale, a novel multi-scale self-attention architecture tailored for document parsing.

1. INTRODUCTION AND PROBLEM STATEMENT

Academic and scientific PDFs present unique challenges due to multi-column formats, mathematical equations, and
Current optical character pipelines and transformer models struggle with boundary detection across contiguous pag
Specifically, existing baselines fail to maintain cross-page continuity, resulting in a 24.3% error rate in multi-page rea
Our primary objective is to resolve spatial degradation while reducing model parameter footprint by 35% without sac

2. METHODOLOGY AND ARCHITECTURE

DocScale incorporates a hierarchical cross-attention mechanism that decouples layout spatial coordinates from lexi
The architecture consists of 12 encoder layers equipped with relative positional bias and gated visual pooling.
By computing cross-attention over multi-scale feature pyramids, the model preserves fine-grained token boundaries

3. DATASET AND EXPERIMENTAL SETUP

We evaluate DocScale on the newly curated SciDoc-50k benchmark, consisting of 50,000 open-access peer-review
The dataset spans four major scientific domains: Computer Science, Physics, Bioengineering, and Computational Li
Data preprocessing involved standard normalization and filtering of documents with fewer than 100 tokens.
All models were trained on 8 NVIDIA A100 GPUs using AdamW optimizer with a learning rate of 2e-4 and cosine de

4. RESULTS AND EMPIRICAL FINDINGS

DocScale achieves a state-of-the-art Macro F1-score of 94.6% on document layout analysis, outperforming previous
Inference throughput reached 142 pages per second on an A100 GPU, reflecting a 3.2x speedup over dense vision
Ablation studies confirm that multi-scale positional gates contribute 2.9% of the total F1 improvement.

5. LIMITATIONS

Despite strong empirical performance, DocScale exhibits several stated limitations.
First, the model performance degrades on historical manuscripts printed prior to 1950 due to font irregularities.
Second, our current implementation assumes single-language English documents and does not natively parse right-
Third, inference memory scales quadratically for documents exceeding 100 pages, necessitating future work in linea

6. CONCLUSION

DocScale provides an efficient, verifiable framework for scientific document summarization and semantic extraction.